Rishabh Pandey

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SUMMARY

AI & Data Science Engineer specializing in building production-ready GenAI applications and predictive ML systems. From developing multi-agent RAG pipelines to deploying scalable FastAPI backends and automated ML workflows, I focus on turning complex data into reliable, high-impact AI solutions.

EDUCATION

Vellore Institute of Technology **2023-2027**
*B.Tech in Computer Science and Engineering (AIML) - CGPA - **8.95/10*** *Bhopal, Madhya Pradesh, India*

SKILLS

Languages: Python, C++, SQL
Generative AI & NLP: LLMs, Transformers, NLP, RAG, Text Embeddings, LangChain, LangGraph, LangSmith, Multi-Agent Systems, HuggingFace
ML & DL: PyTorch, TensorFlow, Scikit-learn, XGBoost, Statsmodels, Time-Series Forecasting
Backend & MLOps: FastAPI, MLflow, Docker, GitHub Actions (CI/CD), SQLAlchemy, Streamlit
Data Analytics & Databases: Pandas, NumPy, Power BI, Plotly, PostgreSQL, MySQL, ChromaDB (VectorDB)
Product & Business: Product Analytics, Product Strategy, User Research, PRDs, User Stories, MVP, A/B Testing

INTERNSHIP

AMASQIS.AI **April 2025 – September 2025**
AI Intern *Remote*

- Built a specialized financial AI model capable of processing complex instructions over **3** training epochs, achieved by fine-tuning DeepSeek-8B using memory-efficient **4-bit QLoRA**.
- Automated a custom data pipeline to cleanly process over **10K+** customer records, and **generated** accurate eligibility predictions across **5** core banking products by establishing a strong XGBoost baseline.
- Delivered highly reliable AI responses by integrating the model into a FastAPI backend, where I implemented strict regex-based guardrails and error fallbacks to safely manage all outputs.

PROJECTS

StorageIQ | Python, PyTorch, Scikit-learn, FastAPI, Docker, GitHub Actions **Jan 2026**

- Predicted **90-day** hardware replacement risks (**VaR_{95} / $CVaR_{95}$**) for **312K+** drives to help optimize capital budgeting, using a **10,000-path** Gamma-Poisson Monte Carlo simulation powered by a PyTorch LSTM ensemble.
- Built a system health anomaly detection framework using Isolation Forests analyzing SMART health telemetry that successfully outperformed traditional baselines across **88** drive models.
- Eliminated runtime inference latency by deploying a FastAPI analytics backend with automated monthly model retraining and GitHub Actions CI/CD for seamless production deployments.

ConvoInsight | Python, Whisper, LangChain, ChromaDB, Groq, yt-dlp, Streamlit **July 2026**

- Engineered an interactive AI meeting assistant that reduces manual review time by automatically extracting summaries and action items from media files using Whisper transcription in a Streamlit UI.
- Built a Retrieval-Augmented Generation (RAG) pipeline using ChromaDB and HuggingFace embeddings, allowing users to instantly search and ask questions across lengthy meeting transcripts.
- Accelerated transcript analysis and query responses by integrating the Groq LLM via LangChain, delivering low-latency, grounded insights directly from processed audio and YouTube videos.

AtlasML | FastAPI, PostgreSQL, Scikit-learn, XGBoost, SQLAlchemy, MLflow, Docker **April 2026**

- Built a full-stack AutoML platform using FastAPI, Scikit-learn, and MLflow automating end-to-end ML workflows from raw data profiling and model recommendation to training, evaluation, and REST API deployment.
- Engineered an automated dataset intelligence engine analyzing critical data-quality signals (class imbalance, missingness, constant features) to compute a **0–100** dataset health score with actionable preprocessing.
- Architected a reproducible production ML infrastructure backed by PostgreSQL, SQLAlchemy, and MLflow to version datasets, track experiment metrics, and seamlessly expose top-performing models for inference.

ACHIEVEMENTS

- Competitive Programming: Achieved a **3-Star** rating on CodeChef and a **1,413** contest rating on LeetCode, solving **350+** Data Structures & Algorithms problems across platforms.
- Kaggle Competitions: Secured a global rank of **1,232** in Kaggle data science competitions through predictive machine learning pipelines and statistical feature engineering.